

Journal-First: Koopman-driven Grip Force Prediction Through EMG Sensing

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Abstract—We study grip force estimation and short-horizon prediction from surface electromyography (sEMG) under deadlines for real-time robotic rehabilitation. Using a single sEMG sensor pair, we process signals in fixed 0.5 s batches and apply Koopman-based models with problem-specific lifting to estimate current grip force and forecast 0.5 s ahead with low weighted mean absolute percentage error (wMAPE). Timing requirements are twofold: (i) a soft setup-time constraint for on-site calibration and model training, and (ii) an online deadline constraint requiring each batch to yield a forecast shortly after arrival. We perform one-time calibration, fit the estimation model in ~ 1.5 s, and measure a maximum observed end-to-end per-batch latency of 27.1 ms—from sEMG preprocessing through estimation, prediction model training, and forecasting—leaving slack for real-time operation. This paper extends the original journal work with an explicit timing specification, deadline interpretation, and a ROS 2 integration status.

Index Terms—Koopman operator theory, electromyography, grip force estimation, robotic rehabilitation

I. PROBLEM STATEMENT AND TIMING REQUIREMENTS

ROBOTIC rehabilitation and assistance can improve safety and interaction quality if the device adapts support to the user *in time*. Grip force is a key control variable: delayed assistance can miss the intent, feel unstable, or fail to provide support during critical phases of a grasp. Surface electromyography (sEMG) provides an early indicator of muscle activation and can therefore enable (i) *grip-force estimation* for feedback and (ii) *short-horizon prediction* for anticipatory control.

Many sEMG-to-grip-force prediction methods improve accuracy by scaling electrode count and relying on offline feature selection, which complicates low-instrumentation deployment. Authors [2] report 2.52% maximum voluntary contraction absolute error using 192-channel high-density sEMG (16 selected) and later show online estimation with eight bipolar electrodes at $\sim 2\%$. These works emphasize accuracy versus windowing and task conditions, but do not report end-to-end periodic timing of the sensing→processing→inference pipeline needed to assess schedulability, so we compare only reported accuracy.

The conference paper is based on [1]. Lab: [ekamenar.github.io](https://github.com/ekamenar). Open-source code: [grip-emg-optimize](#), [shimmer_ros2](#), [godirect_ros2](#), [emg_grip_interfaces](#).

The research is supported by the U.S. Air Force Office of Scientific Research (AFOSR) under Award FA9550-22-1-0531, by the Croatian Science Foundation under grant UIP-2025-02-1541 and by the University of Rijeka under grant uniri-iz-25-116.

We define the following timing requirements: (i) a one-time setup must complete personalized calibration—data collection and model fitting—within a practical on-site timeframe, and (ii) during recurring online operation, each batch must produce a current estimate and short-horizon forecast within a bounded sensing-to-output latency. Setup collects calibration data and fits the estimation model in 1.5 s. The pipeline runs every 0.5 s and outputs a current estimate plus a forecast 0.5 s ahead. This window balances frequency processing with responsiveness. We target an end-to-end batch deadline of 100 ms (consistent with the optimal controller delay in [3]), leaving 400 ms of usable look-ahead.

These timing requirements motivate a pipeline with minimal sensors and bounded computation complexity, combining offline-optimized EMG preprocessing for EMG–force coupling with Koopman models using problem-specific lifting [4]. For nonlinear dynamics $\mathbf{x}_{k+1} = F(\mathbf{x}_k)$, lifting maps the state to features $\phi(\mathbf{x}_k)$, and the Koopman operator $\mathcal{K}$ advances them linearly as $\phi(\mathbf{x}_{k+1}) = \mathcal{K}\phi(\mathbf{x}_k)$. This keeps each batch to FFT preprocessing and compact linear algebra updates for predictable, schedulable runtime.

II. SYSTEM AND METHOD OVERVIEW

We target adaptive assistance for a medium wrap grasp, where sEMG is processed in real time to estimate grip force and predict one batch ahead. We use a single sEMG pair (Shimmer3, ~ 1 kHz) and a calibrated hand dynamometer (Go Direct, 200 Hz, resampled to EMG timestamps).

Data come from a randomized block study with 13 subjects, two electrode positions, five force levels (0–100%), and two repetitions (52 experiments). The spectral mask and smoothing

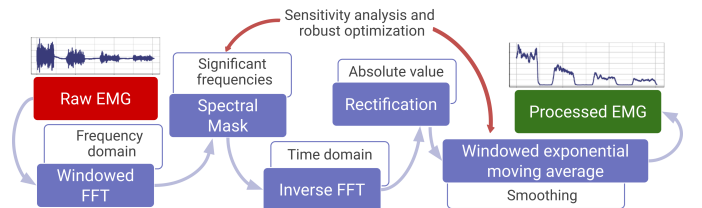


Fig. 1: Signal processing steps for processed EMG optimized for cross-correlation with grip strength [1].

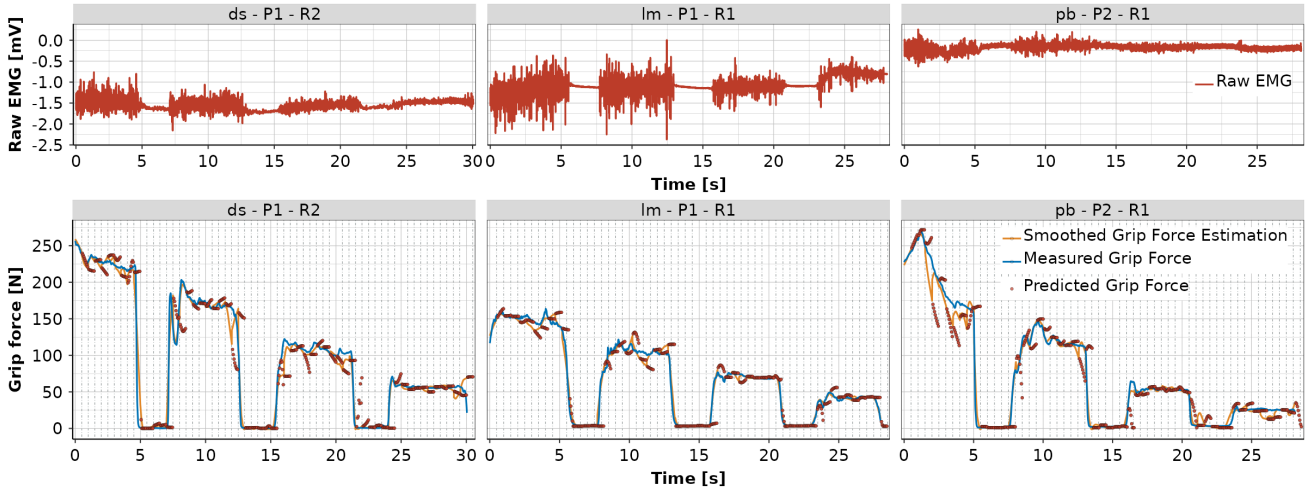


Fig. 2: Examples of raw EMG signals, smoothed estimates, and grip-force predictions [1].

parameters are optimized offline to maximize EMG–force peak cross-correlation (Fig. 1). Raw sEMG is then processed in 0.5 s batches (496 samples) with optimized preprocessing (low-frequency suppression, shaped 20–48 Hz, attenuated 50 Hz, and amplified components up to 202 Hz).

Before each session, and for each user, we record a 20–30 s personalized calibration experiment to fit a Koopman grip force estimator for batch-by-batch online use: the EMG data is preprocessed, then normalized, downsampled 8 times, lifted with time delays and sparse gridded indicator observables, and mapped to grip force via a static Koopman operator [4].

To train the forecast model, we use smoothed force estimation, lift it with time delays and log interaction features, thin the snapshots, and refit a Koopman predictor via QR compressed dynamic mode decomposition (DMD) [5] using hyperparameters selected through grid search. For each batch, the system: (1) acquires sEMG, (2) preprocesses it (Fig. 1), (3) estimates current grip force, and (4) updates the predictor and forecasts 0.5 s ahead, thresholded to the calibrated force range. Maximum observed per-batch times were 0.44 ms for preprocessing, 22.47 ms for estimation, and 4.90 ms for predictor update and forecasting.

Building on the ROS-based setup used in the original work, we are integrating the pipeline in ROS 2. The pipeline up to signal processing is already implemented through sensor drivers, calibration services, preprocessing, and data-collection protocols ([shimmer_ros2](#), [godirect_ros2](#), [emg_grip_interfaces](#)), while estimation and prediction nodes are in progress. Reported end-to-end timings in this paper cover computation only and exclude ROS 2 middleware overhead, which may affect real-time characterization and schedulability in deployment.

III. MAIN RESULTS AND DISCUSSION

Optimized preprocessing yields a strong sEMG–grip-force association (median peak cross-correlation ~ 0.96). Table I summarizes periodic Koopman-driven estimation and 0.5 s-

TABLE I: Summary wMAPE (%) for grip-force estimation and prediction. Position effects are deviations from the mean.

Task	Overall mean	Position 1		Position 2	
		Mean	Effect (pp)	Mean	Effect (pp)
Estimation	5.48	5.7	0.2	5.3	-0.2
Prediction (0.5 s)	17.92	18.0	0.1	17.8	-0.1

ahead prediction wMAPE errors. Electrode position effects are ≤ 0.2 pp and ANOVA confirms they are non-significant.

Setup-time measurements match the intended workflow: calibration takes 20–30 s and estimation model fitting 1.5 s. Online, end-to-end compute per mini-batch averaged 13.5 ms (std. dev. 2.0 ms, 99th percentile 18.6 ms, maximum observed 27.1 ms), leaving more than 72 ms headroom to the 100 ms deadline in the measured runs for OS scheduling and ROS 2 implementation. Fig. 2 shows representative experiment runs.

Together, these results demonstrate a periodic sensing, preprocessing, and forecasting pipeline dominated by FFT preprocessing and small linear-algebra kernels. The framework supports per-session calibration using a single sEMG pair and stable execution under tight control constraints.

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